



Yakeen Plus (2025)

Practice Test- 04

DURATION :200 Minutes

DATE : 09/10/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (4)
2. (4)
3. (1)
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5. (2)
6. (1)
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CHEMISTRY

51. (2)
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ZOOLOGY

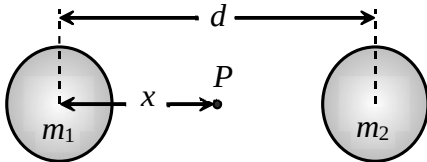
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200. (1)

SECTION-(I) PHYSICS

1. (4)

$$F \propto \frac{1}{r^2}. \text{ If } r \text{ becomes double then } F \text{ reduces to } \frac{F}{4}$$

2. (4)



Force will be zero at the point of zero intensity

$$x = \frac{\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}} d = \frac{\sqrt{81M}}{\sqrt{81M} + \sqrt{M}} D = \frac{9}{10} D$$

3. (1)

$$F \propto xm \times (1-x)m = xm^2(1-x)$$

$$\text{For maximum force } \frac{dF}{dx} = 0$$

$$\Rightarrow \frac{dF}{dx} = m^2 - 2xm^2 = 0 \Rightarrow x = 1/2$$

4. (1)

Gravitational force does not depend on the medium.

5. (2)

The value of g at the height h from the surface of earth

$$g' = g \left(1 - \frac{2h}{R} \right)$$

The value of g at depth x below the surface of earth

$$g' = g \left(1 - \frac{x}{R} \right)$$

These two are given equal,

$$\therefore \left(1 - \frac{2h}{R} \right) = \left(1 - \frac{x}{R} \right)$$

On solving, we get

$$x = 2h$$

6. (1)

$$g = \frac{4}{3} \pi \rho G R. \text{ If } \rho = \text{constant then } \frac{g_1}{g_2} = \frac{R_1}{R_2}$$

7. (2)

Because acceleration due to gravity increases.

8. (2)

 $g = \frac{GM}{R^2}$. If radius shrinks to half of its present value then g will become four times.

9. (3)

$$\Delta U = \frac{mgh}{1 + h/R}$$

Substituting $R = 5h$ we get

$$\Delta U = \frac{mgh}{1 + 1/5} = \frac{5}{6} mgh$$

10. (2)

Dynamic lift = Pressure difference $\times$ Area

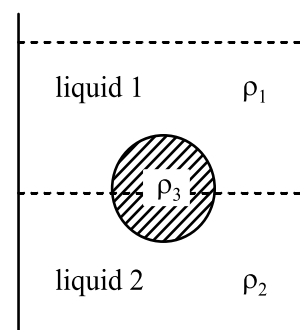
$$= (P_{\text{lower}} - P_{\text{upper}})A$$

According to Bernoulli's theorem

$$= \left(\frac{1}{2} \rho v_{\text{upper}}^2 - \frac{1}{2} \rho v_{\text{lower}}^2 \right) A$$

$$= \frac{1}{2} \rho A \left((2v)^2 - (v)^2 \right) = \frac{3}{2} \rho A v^2$$

11. (1)

The liquid 1 is over liquid 2. Therefore $\rho_1 < \rho_2$.

If ρ_3 had been greater than ρ_2 , it will not be partially inside but anywhere inside liquid 2 if $\rho_3 = \rho_2$ or it would have sunk totally if ρ_3 had been greater than ρ_2 .

$$\therefore \rho_1 < \rho_3 < \rho_2$$

12. (3)

As $\rho_{\text{oil}} < \rho_{\text{water}}$, so oil should be over the water.

As $\rho > \rho_{\text{oil}}$, so the ball will sink in the oil but $\rho < \rho_{\text{water}}$ so it will float in the water.

13. (4)

key: considering the fact that density of coin is very high in comparison of water and wood.

Density of coin is more so it was applying considerable force on wood before falling in water. But once it'll fall, wood can't apply same force alone so wood will come a little up as result l will decrease. when coin falls in water wood goes up so displaced volume decreases and coin has high density so it's volume is low and it doesn't displace liquid that much. As a result h will also decrease.

14. (3)

$$\frac{P_A}{P_B} = \frac{P_0 + \rho gh}{P_0 + \rho gh + 2\rho g(2h)} = \frac{10^5 + 10^5}{10^5 + 5 \times 10^5} = \frac{1}{3}$$

15. (1)

Binding energy = - kinetic energy

And if this amount of energy (E_k) given to satellite then it will escape into outer space

16. (3)

$$\frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2} = \left(\frac{1}{4}\right)^{3/2} = \frac{1}{8}$$

17. (3)

$$l = \frac{FL}{AY}$$

$$\Rightarrow l \propto \frac{L}{d^2}$$

$$\Rightarrow \frac{l_1}{l_2} = \frac{L_1}{L_2} \times \left(\frac{d_2}{d_1}\right)^2$$

$$= \frac{1}{2} \times \left(\frac{1}{2}\right)^2 = \frac{1}{8}$$

18. (2)

$l \propto \frac{1}{r^2}$. If radius of the wire is doubled then

increment in length will become $\frac{1}{4}$ times i.e.,

$$\frac{12}{4} = 3 \text{ mm}$$

19. (4)

$$\text{Energy stored per unit volume} = \frac{1}{2} \times \text{Stress} \times$$

$$\text{Strain} = \frac{1}{2} \times \text{Young's modulus} \times (\text{Strain})^2$$

$$= \frac{1}{2} \times Y \times x^2$$

20. (3)

Remains same

21. (3)

Acceleration due to gravity g varies with dpth as

$$g' = g \left(1 - \frac{d}{R_E}\right)$$

22. (4)

$$K = \frac{\Delta P}{\Delta V/V} = \frac{h\rho g}{\Delta V/V} = \frac{200 \times 10^3 \times 10}{\frac{0.1}{100}} = 2 \times 10^9$$

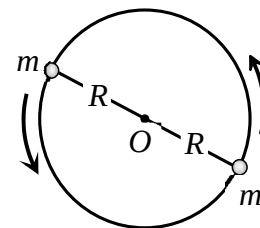
23. (4)

$$\frac{Y_A}{Y_B} = \frac{\tan \theta_A}{\tan \theta_B} = \frac{\tan 60}{\tan 30} = \frac{\sqrt{3}}{1/\sqrt{3}} = 3$$

$$\Rightarrow Y_A = 3Y_B$$

24. (3)

Centripetal force provided by the gravitational force of attraction between two particles



$$\text{i.e., } \frac{mv^2}{R} = \frac{Gm \times m}{(2R)^2}$$

$$\Rightarrow v = \frac{1}{2} \sqrt{\frac{Gm}{R}}$$

25. (4)

$$g = \frac{4}{3} \pi \rho GR$$

$$\therefore \frac{g_1}{g_2} = \frac{R_1 \rho_1}{R_2 \rho_2}$$

26. (3)

$$\frac{v_p}{v_e} = \sqrt{\frac{g_p}{g_e} \times \frac{R_p}{R_e}} = \sqrt{2 \times 2} = 2$$

$$v_p = 2 \times v_e = 2 \times 11.2 = 22.4 \text{ km/s}$$

27. (1)

$$Av = A_1v_1 + A_2v_2 + A_3v_3 + A_4v_4$$

$$Av = 4(A_1v_1)$$

$$12 \times 0.75 = 4[4 \times v]$$

$$v = \frac{3}{4} \times 0.75$$

$$= 0.75 \times 0.75$$

$$v = 0.56 \text{ m/s}$$

28. (1)

$$0.12 \times 3 = 0.12 \times 1.5 + 0.18 \times v$$

$$0.36 = 0.18 + 0.18 \times v$$

$$V = 1 \text{ m/s.}$$

29. (1)

By Bernoulli's principle

30. (1)

The level of water above both points is same so by hydrostatic paradox.

$$p_x = p_y$$

31. (3)

Due to upward acceleration pseudo force will act downwards so value of acceleration due to gravity will increase by 'a'

$$\therefore g' = (g + a)$$

$$P = \rho g' h$$

$$\Rightarrow P = \rho(g + a)h \quad (\text{Substitute } g')$$

32. (2)

Pressure in a liquid is divided equally so we can say pressure at both the pistons should be same

$$\Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \left\{ \begin{array}{l} \text{Where,} \\ F_1 = F \\ A_1 = 10 \text{ cm}^2 \\ A_2 = 10 \text{ m}^2 = 10 \times 10^4 \text{ cm}^2 \\ F_2 = 8000 \text{ N} \end{array} \right.$$

Substituting values,

$$\frac{F}{10} = \frac{8000}{10 \times 10^4}$$

$$\Rightarrow F = 0.8 \text{ N} \quad \left\{ \begin{array}{l} \text{Take} \\ g = 10 \text{ m/s}^2 \end{array} \right.$$

33. (3)

We know that, Bulk modulus

$$B = \frac{-dP}{\frac{dV}{V}}$$

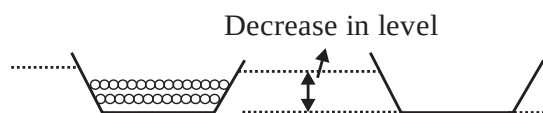
As liquid is incompressible, $dV = 0$

$$\Rightarrow \beta = \frac{-dP}{0} = \infty$$

So, Bulk modulus for an incompressible liquid is infinity.

34. (3)

Previously when stones are on the boat they are increasing the weight on the boat and to balance this weight boat needs to generate buoyancy force by displacing more water, but when stones are removed the boat starts displacing less amount of water hence the level of water in tank falls.



35. (4)

Let each have mass = M and densities d_1 and d_2

$$d_{\text{mix}} = \frac{M_{\text{mix}}}{V_{\text{mix}}} = \frac{M + M}{\left(\frac{M}{d_1}\right) + \left(\frac{M}{d_2}\right)} = \frac{2d_1d_2}{d_1 + d_2}$$

36. (1)

Velocity of efflux for small holes = $\sqrt{2gh}$

Which clearly is independent of ' ρ ' (density)

So, $v_1 = v_2$

37. (1)

Pressure close to the base = ρgH

Force required = pressure $\times$ area of hole = $\rho gH(\pi r^2)$

38. (2)

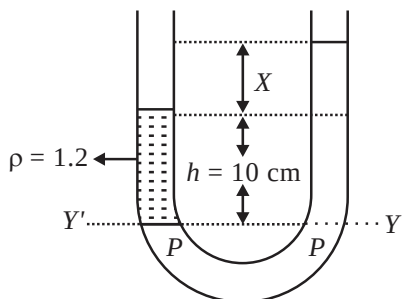
Let the difference between 2 limbs be x

Pressure on the line $Y Y'$ should be same below both limbs, so

$$\rho_{\text{glycerine}} \times g \times h = \rho_{\text{water}} \times g \times (h + x)$$

$$\Rightarrow 1.2 \times 10 = 1 \times (h + x)$$

$$\Rightarrow 2 \text{ cm} = x$$



39. (3)

Let height of water in tank be h

$$\text{So, } 4P - P = \rho_w gh \quad \dots (1)$$

$\therefore \frac{3}{5}$ water taken out $\frac{2}{5}$ th water is left to exert pressure

$$P' = P + \frac{2}{5} \rho_w gh$$

$$\Rightarrow P' = P + \frac{2}{5} \times 3P \quad [\text{From eq. (1)}]$$

$$\Rightarrow P' = \frac{11P}{5}$$

40. (4)

$$\text{Gravitational potential} = \int E dx = \int_x^\infty \frac{K}{x^3} dx$$

$$= K \left(\frac{x^{-3+1}}{-3+1} \right)_x^\infty = \left[\frac{-K}{2x^2} \right]_x^\infty = \frac{K}{2x^2}$$

41. (1)

$$K.E. = \frac{GMm}{2R}$$

42. (2)

$$v_e = \sqrt{\frac{2GM}{R}} = R \sqrt{\frac{8}{3} \pi G \rho}$$

If mean density is constant then $v_e \propto R$

$$\frac{v_e}{v_p} = \frac{R_e}{R_p} = \frac{1}{2}$$

$$\Rightarrow v_e = \frac{v_p}{2}$$

43. (4)

$$v_0 = \sqrt{\frac{GM}{r}}$$

44. (1)

Escape velocity is same for all angles of projection.

45. (3)

$$\frac{F_A}{F_B} = \frac{P_A A_A}{P_B A_B} = \frac{\rho_A g h_A A_A}{\rho_B g h_B A_B} = 1$$

46. (4)

$$A_A v_A = A_C v_C$$

$$\text{If } A_R = A_C \text{ then } v_A = v_C$$

$$\therefore P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$$

if P , v and ρ is same then h will also same.

47. (4)

$$L = mvr = m \sqrt{\frac{GM}{r}} r = m \sqrt{GM} r$$

$$\therefore L \propto \sqrt{r}$$

48. (1)

$$\text{Energy per unit volume} = \frac{1}{2} \times Y \times (\text{Strain})^2$$

$$\therefore \text{Strain} = \sqrt{\frac{2E}{Y}}$$

49. (1)

$$\text{From the graph } l = 10^{-4} \text{ m, } F = 20 \text{ N}$$

$$A = 10^{-6} \text{ m}^2, L = 1 \text{ m}$$

$$\therefore Y = \frac{FL}{Al}$$

$$= \frac{20 \times 1}{10^{-6} \times 10^{-4}} = 20 \times 10^{10} = 2 \times 10^{11} \text{ N/m}^2$$

50. (4)

$$\text{Volume flowing per second} = Av$$

$$= A \sqrt{2gh} = 10 \times 10^{-4} \sqrt{2 \times 10 \times 5} = 10^{-2} \text{ m}^3/\text{s}$$

SECTION-(II) CHEMISTRY

51. (2)

Greater are the concentrations of the reactants, faster is the reaction. Conversely, as the concentrations of the reactants decrease, the rate of reaction also decreases.

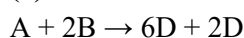
[New NCERT Class 12th Page No. 63]

52. (4)

The rate of a reaction depends upon concentration of reactant, pressure (for gaseous reactions) and temperature.

[New NCERT Class 12th Page No. 62]

53. (3)



$$-\frac{\Delta[A]}{\Delta t} = \frac{1}{2} \left(-\frac{\Delta[B]}{\Delta t} \right) = \frac{1}{6} \left(\frac{\Delta[C]}{\Delta t} \right) = \frac{1}{2} \left(\frac{\Delta[D]}{\Delta t} \right)$$

$$-\frac{\Delta[A]}{\Delta t} = \frac{1}{2} \left(-\frac{\Delta[B]}{\Delta t} \right)$$

$$-2.6 \times 10^{-2} = \frac{1}{2} \left(-\frac{\Delta[B]}{\Delta t} \right)$$

$$-\frac{\Delta[B]}{\Delta t} = 5.2 \times 10^{-2} \text{ Ms}^{-1}$$

[New NCERT Class 12th Page No. 65]

54. (2)

The resistance decreases with increase in temperature.

[New NCERT Class 12th Page No. 39]

55. (1)

$$\kappa = \frac{\text{Cell constant}}{R} = \frac{0.5}{50} = 0.01$$

$$\Lambda_{\text{eq}} = \frac{\kappa \times 1000}{N} = \frac{0.01 \times 1000}{N} = \frac{1}{100} \times \frac{1000}{1}$$

$$\Lambda_{\text{eq}} = 10 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

[New NCERT Class 12th Page No. 44]

56. (4)

This is due to increase in the degree of ionization of the electrolyte.

[New NCERT Class 12th Page No. 39]

57. (1)

$$\Lambda_m = \frac{\kappa \times 1000}{M}$$

$\therefore \Lambda_m$ is inversely proportional to molarity.

[New NCERT Class 12th Page No. 44]

58. (4)

Order of reaction may be positive, negative, zero or fractional.

[New NCERT Class 12th Page No. 70]

59. (4)

$$t_{1/2} = 138 \text{ sec}$$

$$1.28 \xrightarrow{t_{1/2}} 0.64 \xrightarrow{t_{1/2}} 0.32 \xrightarrow{t_{1/2}} 0.16 \xrightarrow{t_{1/2}} 0.08 \xrightarrow{t_{1/2}} 0.04$$

$$\text{Time required} = 5t_{1/2} = 5 \times 138$$

$$= 690 \text{ sec}$$

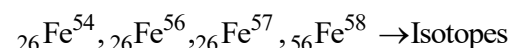
[New NCERT Class 12th Page No. 70]

60. (2)

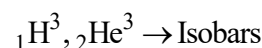
The mass and the positive charge of an atom are concentrated in the nucleus.

[New NCERT Class 11th Page No. 37]

61. (1)



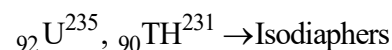
having same atomic number but different mass number



having difference in number of electron, proton and neutron but sum of number of neutrons and protons remain same.



they are atoms of different element which have the same number of neutrons.



Having the same difference of the number of neutron and protons.

[New NCERT Class 11th Page No. 35]

62. (2)

$$\text{As we know, } \bar{v} = \frac{1}{\lambda} = \frac{1}{4600 \text{ \AA}}$$

$$= \frac{1}{4600 \times 10^{-8} \text{ cm}}$$

$$\frac{10^8}{4600} \text{ cm}^{-1} = 21739 \text{ cm}^{-1}$$

[New NCERT Class 11th Page No. 33]

63. (2)

$$\begin{array}{lcl} & A \rightarrow B & \\ t = 0 & 5 & 0 \\ t = 20 \text{ min} & 5 - x & x \\ A_0 - A_t = kt & & \\ A_0 - (A_0 - x) = kt & & \\ \therefore x = kt = [B] & & \\ \Rightarrow [B] = 0.6 \times 10^{-3} \times 20 \times 60 & & \\ = 0.72 \text{ M} & & \end{array}$$

[New NCERT Class 12th Page No. 71]

64. (4)

For $n = 3$, $\ell = 2 \Rightarrow m = -3$ not possible.

[New NCERT Class 11th Page No. 36]

65. (2)

$$\begin{array}{l} E \propto \frac{1}{\lambda} \\ \frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} \Rightarrow \frac{25}{50} = \frac{\lambda_2}{\lambda_1} \Rightarrow \lambda_1 = 2\lambda_2 \end{array}$$

[New NCERT Class 11th Page No. 44]

66. (1)

In general,

$$t_{1/2} \propto (a)^{1-n}$$

n = order of reaction

[New NCERT Class 12th Page No. 73]

67. (4)

$$c = v\lambda$$

$$\lambda = \frac{c}{v} = \frac{3 \times 10^{17}}{6 \times 10^{15}} = 50 \text{ nm}$$

[New NCERT Class 11th Page No. 41]

68. (1)

$$\begin{array}{l} \Lambda_m = \frac{\kappa \times 1000}{M} \\ = \frac{6.3 \times 10^{-2} \times 1000}{0.1} \\ = 630 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1} \end{array}$$

[New NCERT Class 12th Page No. 35]

69. (1)

$$\text{Rate of Reaction} = -\frac{d[N_2]}{dt} = -\frac{1}{3} \frac{d[H_2]}{dt} = \frac{1}{2} \frac{d[NH_3]}{dt}$$

[New NCERT Class 12th Page No. 63]

70. (3)

$$r = k [A]^x [B]^y \text{ ---(1)}$$

$$A' = 2A, B' = B \text{ and } r' = 2r$$

$$\therefore r' = k.(A')^x (B')^y$$

$$\text{Hence } 2r = k.(2A)^x.(B)^y \text{ ---(2)}$$

Now (2)/(1)

$$\frac{2r}{r} = (2)^x$$

$$x = 1$$

$$\text{Again } r'' = 8r, A'' = 2A \text{ and } B'' = 2B$$

$$r'' = k.(A'')^x (B'')^y$$

$$\text{or } 8r = k.2^x(A)^x 2^y(B)^y$$

$$8r = 2^x.2^y.k(A)^x (B)^y \text{ --- (3)}$$

From (3)/(1)

$$\frac{8r}{r} = 2^y$$

$$\text{or } y = 2$$

$$\therefore r = [A]^1 [B]^2$$

[New NCERT Class 12th Page No. 56]

71. (4)

$$t_{1/2} = 1386 \text{ sec}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{1386}$$

$$k = 0.5 \times 10^{-3} \text{ s}^{-1}$$

[New NCERT Class 12th Page No. 60]

72. (4)

For endothermic reaction $E_a > \Delta H$

[New NCERT Class 12th Page No. 65]

73. (2)

$$\kappa = 0.012 \text{ ohm}^{-1} \text{ cm}^{-1}$$

$$R = 55 \text{ ohm}$$

$$G = \frac{1}{55} \text{ ohm}^{-1}$$

$$\text{Cell constant} = \frac{\kappa}{G} = \frac{0.012}{1/55} = 0.012 \times 55 = 0.66 \text{ cm}^{-1}$$

[New NCERT Class 12th Page No. 45]

74. (2)

$$\Delta G^\circ = -nFE^\circ$$

$$\Delta G_3^\circ = \Delta G_1^\circ - \Delta G_2^\circ$$

$$\Rightarrow E_3^\circ = (2 \times E_1^\circ) - E_2^\circ$$

$$= (2 \times 0.337) - 0.153$$

$$= 0.52 \text{ V}$$

[New NCERT Class 12th Page No. 41]

75. (3)

$$\therefore r = 0.529 \times \frac{n^2}{Z} \text{ \AA} \quad (Z=3, \text{ for Li})$$

$$\therefore r_2 = 0.529 \times \frac{(2)^2}{3} = 0.529 \times \frac{4}{3}$$

$$r_4 = 0.529 \times \frac{(4)^2}{3} = 0.529 \times \frac{16}{3}$$

$$\text{Therefore } \frac{r_2}{r_4} = \frac{0.529 \times \frac{(2)^2}{3}}{0.529 \times \frac{(4)^2}{3}} = \frac{1}{4}$$

$$\text{or } r_2 : r_4 = 1 : 4$$

[New NCERT Class 11th Page No. 32]

76. (3)

$$\text{Angular momentum} = \sqrt{\ell(\ell+1)} \frac{h}{2\pi}$$

for p-subshell, $\ell = 1$

$$= \sqrt{1(1+1)} \frac{h}{2\pi} = \frac{\sqrt{2}h}{2\pi}$$

[New NCERT Class 11th Page No. 31]

77. (2)

$$\text{time taken} = \frac{\text{distance}}{\text{velocity}}$$

$$= \frac{2\pi r}{v} = \frac{2 \times 3.14 \times 0.529 \times \frac{(3)^2}{3} \times 10^{-10} \text{ m}}{2.18 \times 10^6 \times \frac{3}{3} \text{ ms}^{-1}} = 4.57 \times 10^{-16} \text{ s}$$

[New NCERT Class 11th Page No. 34]

78. (3)

protons (p) = 17

electrons (e⁻) = 18

neutrons (n) = 18

for neutral atoms

number of protons = number of electrons

Here number of e⁻ > number of (p)

So charge is -1

[New NCERT Class 11th Page No. 33]

79. (3)

At equilibrium, $Q = K_{eq}$ and $E_{cell} = 0$

$$E_{cell} = E_{cell}^{\circ} - \frac{0.059}{1} \log K_{eq}$$

$$\Rightarrow \log K_{eq} = \frac{0.59}{0.059} = 10$$

$$K_{eq} = 1.0 \times 10^{10}$$

[New NCERT Class 12th Page No. 46]

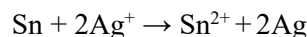
80. (1)

According to electro chemical series.

$K > Mg > Zn > Cu$

[New NCERT Class 12th Page No. 44]

81. (1)



$$E_{cell} = E_{cell}^{\circ} - \frac{0.0591}{2} \log \frac{[Sn^{2+}]}{[Ag^+]^2}$$

On increasing the concentration of Ag⁺ ion, the value of cell potential comes out to be more positive.

[New NCERT Class 12th Page No. 45]

82. (3)

$${}^{16}_8\text{O} \rightarrow \left[\begin{array}{l} n = 8 \text{ amu} \\ p = 8 \text{ amu} \end{array} \right] = 16 \text{ amu}$$

If mass of neutron is halved, mass of proton is doubled and mass of electron is 4 times, then new atomic mass will be:

$$\left[\begin{array}{l} n = 4 \text{ amu} \\ p = 16 \text{ amu} \end{array} \right] = 20 \text{ amu}$$

$$\% \text{ Increment} = \frac{\text{Final mass} - \text{Initial mass}}{\text{Initial mass}} \times 100$$

$$= \frac{20 - 16}{16} \times 100 = 25\%$$

[New NCERT Class 11th Page No. 33]

83. (2)

$$K = Ae^{-E_a/RT}$$

$$\ln K = \ln A - \frac{E_a}{RT}$$

$$\ln K_1 = \ln A - \frac{E_a}{RT_1} \quad \dots(1)$$

$$\ln K_2 = \ln A - \frac{E_a}{RT_2} \quad \dots(2)$$

Equation (2) – Equation (1)

$$\ln K_2 - \ln K_1 = -\frac{E_a}{RT_2} + \frac{E_a}{RT_1}$$

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

[New NCERT Class 12th Page No. 69]

84. (4)

$$\Delta H = (E_a)_f - (E_a)_b$$

$$\Delta H = 0$$

[New NCERT Class 12th Page No. 71]

85. (1)
For first order reaction
$$t_{1/2} = \frac{0.693}{k}$$

[New NCERT Class 12th Page No. 69]

86. (3)
Electromagnetic radiation travels through vacuum at a speed of light = $3 \times 10^8 \text{ ms}^{-1}$
[New NCERT Class 11th Page No. 41]

87. (2)
 $\therefore 1\text{eV} = 1.602 \times 10^{-19} \text{ J}$
 $\therefore 5\text{eV} = 8 \times 10^{-19} \text{ J} = E \text{ (Energy)}$
Calculation of wavelength (λ):
$$E = \frac{hc}{\lambda} \text{ or } \lambda = \frac{hc}{E}$$

$$\lambda = \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{8 \times 10^{-19} \text{ J}}$$

$$\lambda = 2.5 \times 10^{-7} \text{ m}$$

[New NCERT Class 11th Page No. 41]

88. (3)
EMF of a cell = Reduction potential of cathode – Reduction potential of anode or Reduction potential of cathode + oxidation potential of anode
Reduction potential = –(oxidation potential)
[New NCERT Class 12th Page No. 37]

89. (4)
 $\Delta G^\circ = -nFE_{\text{cell}}^\circ$
If $E_{\text{cell}}^\circ = -\text{ve}$ then $\Delta G^\circ = +\text{ve}$ i.e. $\Delta G^\circ > 0$
 $\Delta G^\circ = -nRT \log K_{\text{eq}}$
For $\Delta G^\circ = +\text{ve}$; $K_{\text{eq}} = -\text{ve}$ i.e. $K_{\text{eq}} < 1$
[New NCERT Class 12th Page No. 45]

90. (2)
 $\Delta_{\text{CH}_3\text{COOH}}^\circ = \Delta_{\text{CH}_3\text{COONa}}^\circ + \Delta_{\text{HCl}}^\circ - \Delta_{\text{NaCl}}^\circ$
 $= 91.0 + 425.9 - 126.4$
 $= 390.5 \text{ S cm}^2 \text{ mol}^{-1}$
[New NCERT Class 12th Page No. 47]

91. (4)
 $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
$$w = \frac{E}{96500} \times I \times t$$

moles of Cl_2 gas $\times$ molecular weight of Cl_2
 $= \frac{\text{Equivalent weight of } \text{Cl}_2 \text{ gas} \times I \times t}{96500}$
$$\Rightarrow 0.1 \times 71 = \frac{35.5}{96500} \times 3 \times t$$

$$\Rightarrow t = 6433.33 \text{ sec} = 107 \text{ min}$$

[New NCERT Class 12th Page No. 45]

92. (2)
 $n = 4$ to $n = 2$
Visible region (green)
[New NCERT Class 11th Page No. 31]

93. (1)
$$\Delta x \times \Delta p = \frac{h}{4\pi} = \text{constant}$$

 $\therefore \Delta p \text{ of helium atom} = 32 \times 10^5 \text{ kg ms}^{-1}$
[New NCERT Class 11th Page No. 39]

94. (1)
For A, $n = 3$, $\ell = 2 \Rightarrow 3d$
For B, $n = 5$, $\ell = 0 \Rightarrow 5s$
They energy of orbital 'B' is more than orbital 'A'
[New NCERT Class 11th Page No. 36]

95. (2)
Maximum number of electrons in a subshell
 $= 2(2l + 1)$
 $= 4l + 2$
[New NCERT Class 11th Page No. 40]

96. (4)
For a first order reaction; $t_{1/2} = \frac{0.693}{k}$
$$t_{1/2} = \frac{0.693}{2.303 \times 10^{-3}} = 301\text{s}$$

The time required for 40 g of reactant to reduce to 10 g
 $t_{75\%} = 2 \times t_{1/2}$
 $t_{75\%} = 2 \times 301 = 602 \text{ s}$
[New NCERT Class 12th Page No. 65]

97. (1)
Half life is the time for half of the substance of decay.
[New NCERT Class 12th Page No. 70]

98. (4)
HCl is strong electrolyte as it completely dissociates when dissolved in water.
[New NCERT Class 12th Page No. 33]

99. (4)
Two electrons occupying the same orbital are distinguished by spin quantum number.
[New NCERT Class 11th Page No. 32]

100. (3)
Angular nodes = $\ell = 3$
Total nodes = $n - 1 = 3$
 $n = 4$
 $\therefore 4f \text{ orbital}$
[New NCERT Class 11th Page No. 36]

SECTION-(III) BOTANY

- | | |
|--|--|
| <p>101. (1)
Blue-green algae, also known as cyanobacteria have a mucilagenous sheath, which is a slimy protective layer that surrounds the cells.
[NEW NCERT Class 11th Page No.13]</p> <p>102. (2)
Viruses and viroids are acellular organisms and are not included in any of the five kingdoms classified by Whittaker.
[NEW NCERT Class 11th Page No. 19]</p> <p>103. (2)
<i>Penicillium</i> is multicellular and are source of antibiotics.
[NEW NCERT Class 11th Page No.16]</p> <p>104. (1)
Bacteria actually show great metabolic diversity, with various types of bacteria capable of different metabolic processes, including photosynthesis, nitrogen fixation, and chemosynthesis.
[NEW NCERT Class 11th Page No. 13]</p> <p>105. (2)
Spores of slime moulds are dispersed by air currents.
[NEW NCERT Class 11th Page No. 15]</p> <p>106. (3)
The pigments of euglenoids are identical to those present in higher plants.
[NEW NCERT Class 11th Page No.15]</p> <p>107. (3)
 <ul style="list-style-type: none"> • In general, viruses that infect plants have single stranded RNA • Viruses contain either RNA or DNA. [NEW NCERT Class 11th Page No.20]</p> <p>108. (3)
Heterocyst present in <i>Anabaena</i> is specialised for nitrogen fixation.
[NEW NCERT Class 11th Page No. 13]</p> <p>109. (4)
Diatoms and golden algae (desmids) are found in fresh water as well as in marine environments.
[NEW NCERT Class 11th Page No. 14]</p> <p>110. (3)
Spores of slime moulds possess <u>true walls</u> and survive for many years under <u>adverse conditions</u>.
[NEW NCERT Class 11th Page No. 15]</p> | <p>111. (1)
Deuteromycetes are commonly known as imperfect fungi because only the asexual or vegetative phases of these fungi are known.
[NEW NCERT Class 11th Page No. 18]</p> <p>112. (2)
 <ul style="list-style-type: none"> • Archaeobacteria are known for tolerating extreme conditions, including high temperatures (thermophiles). • Archaeobacteria, particularly halophiles, are found in extreme saline environments. • Some archaeobacteria, like methanogens, are involved in the production of biogas. [NEW NCERT Class 11th Page No. 13]</p> <p>113. (3)
 <ul style="list-style-type: none"> • The cell walls of fungi are composed of chitin and polysaccharides. • Most fungi are saprophytic in their mode of nutrition. [NEW NCERT Class 11th Page No. 16]</p> <p>114. (2)
The kingdom Plantae includes eukaryotic organisms.
[NEW NCERT Class 11th Page No. 19]</p> <p>115. (4)
<i>Neurospora</i> and <i>Penicillium</i> form exogenous asexual spores (conidia) and endogenous sexual spores (ascospores).
[NEW NCERT Class 11th Page No. 18]</p> <p>116. (2)
The cyanobacteria (also referred to as blue-green algae) have chlorophyll <i>a</i> similar to green plants and are photosynthetic autotrophs.
[NEW NCERT Class 11th Page No. 13]</p> <p>117. (4)
The sex organs are absent in the members of basidiomycetes.
[NEW NCERT Class 11th Page No. 18]</p> <p>118. (4)
The boundaries of Kingdom Protista cannot be well-defined because this kingdom forms a link with the others dealing with fungi, plants animals. Hence, Both A and R are true but R is NOT the correct explanation of A.
[NEW NCERT Class 11th Page No. 14]</p> |
|--|--|

119. (4)
Cyanobacteria often form blooms in polluted water bodies.

[NEW NCERT Class 11th Page No. 13]

120. (3)
Kingdom Protista includes all single-celled eukaryotes.

[NEW NCERT Class 11th Page No. 15]

121. (3)
- Heterotrophic bacteria have a significant impact on human affairs.
 - Chemosynthetic autotrophic bacteria play a great role in recycling nutrients like nitrogen, phosphorous and sulphur.

[NEW NCERT Class 11th Page No. 13]

122. (4)
In the life cycle of the protozoan *Plasmodium*, which causes malaria, an infectious spore-like stage called sporozoite is present.

[NEW NCERT Class 11th Page No. 16]

123. (4)
Bacterial viruses or bacteriophages infect the bacteria.

[NEW NCERT Class 11th Page No. 20]

124. (3)
Commonly known forms of basidiomycetes are mushrooms, bracket fungi or puffballs.

[NEW NCERT Class 11th Page No.18]

125. (4)
- | | |
|-----------------|---|
| Dinoflagellates | Releases toxins that kill marine animals |
| Protozoans | Primitive relatives of animals |
| Diatoms | Cell wall contain silica |
| Euglenoids | In absence of sunlight, they behave like heterotrophs |

[NEW NCERT Class 11th Page No. 15-16]

126. (1)
Under suitable conditions, slime mould form an aggregation called plasmodium which may grow and spread over several feet.

[NEW NCERT Class 11th Page No. 15]

127. (1)
The Mycoplasma are eubacteria that completely lack a cell wall. They are the smallest living cells known and can survive without oxygen. Many mycoplasmas are pathogenic in animals and plants.

[NEW NCERT Class 11th Page No. 13-14]

128. (1)
Diatoms are the chief producers in the oceans.

[NEW NCERT Class 11th Page No. 14]

129. (3)
Slime moulds are saprophytic protists. The body moves along decaying twigs and leaves engulfing organic material. Hence, reason R explains why slime moulds are classified as saprophytic protists, making both statements true and the reason an appropriate explanation for the assertion.

[NEW NCERT Class 11th Page No. 15]

130. (1)
Fusion of protoplasts between two motile or non-motile gametes called plasmogamy.

[NEW NCERT Class 11th Page No. 17]

131. (3)
Gonyaulax undergo such rapid multiplication that they make the sea appear red (red tides).

[NEW NCERT Class 11th Page No. 15]

132. (4)
- | | |
|--------------------|-----------------|
| <i>Aspergillus</i> | Ascospore |
| <i>Mucor</i> | Zygospore |
| <i>Puccinia</i> | Basidiospore |
| <i>Trichoderma</i> | Imperfect fungi |

[NEW NCERT Class 11th Page No. 17-18]

133. (2)
In *Mucor*, *Rhizopus* and *Albugo* the mycelium is aseptate and coenocytic.

[NEW NCERT Class 11th Page No. 17]

134. (3)
In Linnaeus' time a two Kingdom system of classification with Plantae and Animalia kingdoms was developed that included all plants and animals respectively. This system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organisms.

[NEW NCERT Class 11th Page No. 10]

135. (4)
- *Alternaria* belongs to imperfect fungi.
 - *Agaricus* belongs to basidiomycetes.
 - *Albugo* belongs to phycomycetes.

[NEW NCERT Class 11th Page No. 17-18]

136. (2)
Prions was similar in size to viruses.

[NEW NCERT Class 11th Page No. 20-21]

137. (3)

The imperfect fungi which are decomposers of litter and help in mineral cycling belong to deuteromycetes.

[NEW NCERT Class 11th Page No. 18]

138. (4)

Bladderwort is a parasitic plant.

[NEW NCERT Class 11th Page No.14]

139. (2)

The bacterial structure is very simple but they are very complex in behaviour.

[NEW NCERT Class 11th Page No.12-13]

140. (2)

The most notorious sporozoan is *Plasmodium* (malarial parasite) which causes malaria, a disease which has a staggering effect on human population.

[NEW NCERT Class 11th Page No. 16]

141. (3)

Viruses cause diseases like mumps, small pox, herpes and influenza.

[NEW NCERT Class 11th Page No. 14, 20]

142. (4)

Heterotrophic bacteria are most abundant in nature.

[NEW NCERT Class 11th Page No. 13]

143. (3)

- Heterotrophic bacteria such as *Lactobacillus* helpful in making curd from milk.
- Archaeobacteria and eubacteria having a different cell wall structure.

[NEW NCERT Class 11th Page No. 13]

144. (1)

Karyogamy and meiosis take place in the basidium producing four basidiospores. The basidiospores are exogenously produced on the basidium.

[NEW NCERT Class 11th Page No.18]

145. (1)

<i>Rhizopus</i>	Bread mould
Prions	Causes mad cow disease in cattle
Animalia	Organ system level of body organisation
Viroids	Causes potato spindle tuber disease

[NEW NCERT Class 11th Page No.11,21]

146. (2)

The morphology of the mycelium, mode of spore formation and fruiting bodies form the basis for the division of the kingdom into various classes.

[NEW NCERT Class 11th Page No. 17]

147. (1)

In lichen, algae prepare food for fungi and fungi provide shelter and absorb mineral nutrients and water for its partner.

[NEW NCERT Class 11th Page No. 21]

148. (2)

- T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease.
- It was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid.

[NEW NCERT Class 11th Page No. 21]

149. (1)

Bladderwort and Venus fly trap are examples of insectivorous plants and *Cuscuta* is a parasite

[NEW NCERT Class 11th Page No. 19]

150. (2)

Diatoms cell walls are embedded with silica along with cellulose.

Dinoflagellates cell walls are composed of cellulose, arranged in plates.

[NEW NCERT Class 11th Page No. 14-15]

SECTION-(IV) ZOOLOGY

151. (2)

The amount of filtrate formed by the kidney per minute is 125 ml. This is called glomerular filtration rate.

[New NCERT Class 11th Page No. 240]

152. (4)

The glomerular capillary blood pressure causes filtration of blood through 3 layers, i.e., the endothelium of glomerular blood vessels, the epithelium of Bowman's capsule and a basement substance between these two layers.

[Old NCERT Class 11th Page No. 293]

153. (4)

Ammonotelic	Bony fishes
Uricotelic	Reptiles
Ureotelic	Marine fishes
Bowman's capsule	Renal tubule

[New NCERT Class 11th Page No. 242]

154. (2)

In prawn, antennal gland helps in the removal of nitrogenous waste.

[New NCERT Class 11th Page No. 207]

155. (3)

- Human kidneys can produce urine nearly four times concentrated than the initial filtrate formed.
- The functioning of the kidneys is efficiently monitored and regulated by hormonal feedback mechanisms involving the hypothalamus, JGA and to a certain extent, the heart.

[New NCERT Class 11th Page No. 212]

156. (3)

The renal cortex extends between the medullary pyramid as columns of bertini.

[New NCERT Class 11th Page No. 207]

157. (3)

Flatworm	Flame cell
Land snail	Excrete uric acid
<i>Periplaneta</i>	Malpighian tubule
Mammals	Excrete urea

[New NCERT Class 11th Page No. 214]

158. (1)

In PCT, nearly all of the essential nutrients and 70-80 percent of electrolytes and water are reabsorbed in which segment of nephron.

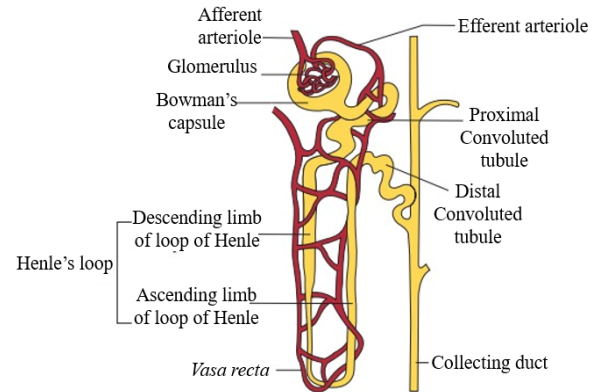
[New NCERT Class 11th Page No. 209]

159. (3)

During urine formation, the tubular cells secrete substances like H^+ , K^+ and ammonia into the filtrate. Tubular secretion is also an important step in urine formation as it helps in the maintenance of ionic and acid base balance of body fluids.

[New NCERT Class 11th Page No. 209]

160. (4)



In collecting duct of nephron the filtrate is called urine.

[New NCERT Class 11th Page No. 207]

161. (3)

Presence of glucose and ketone bodies in urine are indicative of diabetes mellitus.

[New NCERT Class 11th Page No. 213]

162. (2)

Ascending limb of loop of Henle is impermeable to water and allow transport of electrolyte actively and passively.

[New NCERT Class 11th Page No. 244]

163. (1)

200 ml CO_2 removed by our lungs per minute.

[New NCERT Class 11th Page No. 214]

164. (4)

Kidneys are reddish brown, bean shaped structures situated between the levels of last thoracic and third lumbar vertebra close to the dorsal inner wall of the abdominal cavity.

[New NCERT Class 11th Page No. 206]

165. (2)

Podocyte of Bowman's capsule formed the filtration slit.

[New NCERT Class 11th Page No. 208]

166. (3)

Vasa recta is well developed in juxta medullary nephrons.

[New NCERT Class 11th Page No. 208]

167. (4)

In response, the stretch receptors on the walls of the bladder send signals to the CNS. The CNS passes on motor messages to initiate the contraction of smooth muscles of the bladder and simultaneous relaxation of the urethral sphincter causing the release of urine.

[New NCERT Class 11th Page No. 212]

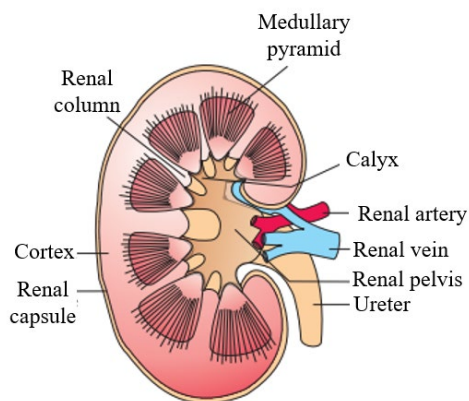
168. (1) The medullary interstitial fluid has a very high osmolarity which is maintained by the movement of urea from collecting duct into medullary interstitium. [New NCERT Class 11th Page No. 209] 169. (1) Renin release from JG cell of juxtaglomerular apparatus is in response to fall in glomerular filtration rate. [New NCERT Class 11th Page No. 209] 170. (2) Urine formation take place in the absence of loop of Henle. [New NCERT Class 11th Page No. 209] 171. (4) • Bowman's capsule is double walled cup-like structure. • Glomerulus alongwith Bowman's capsule, is called the malpighian body or renal corpuscle. • Elimination of urea requires the least amount of water. • Urine formation involves filtration, reabsorption and secretion. [New NCERT Class 11th Page No. 206] 172. (4) Presence of peritubular capillaries can not be considered as point of difference between cortical and juxtamedullary nephrons. [New NCERT Class 11th Page No. 210] 173. (4) Angiotensin-II increase reabsorption of Na⁺ from the distal part of tubules. [New NCERT Class 11th Page No. 212] 174. (3) Glomerulus is not considered as component of tubular part of nephron. [New NCERT Class 11th Page No. 214] 175. (2) Under the influence of ADH maximum reabsorption of water take place in collecting duct of nephron. [New NCERT Class 11th Page No. 214] 176. (4) Albumin is absent in filtrate. [New NCERT Class 11th Page No. 210] 177. (3) Hemodialysis is a medical procedure that filters waste products, including urea, from the blood when the kidneys are unable to do so. [Old NCERT Class 11th Page No. 298]	178. (3) PCT is lined by simple cuboidal brush border epithelium which increases the surface area for reabsorption. [Old NCERT Class 11th Page No. 294] 179. (3) Each human kidney has one million complex tubular structures called nephrons. [New NCERT Class 11th Page No. 207] 180. (4) Periubular capillaries of nephron arise from efferent arterioles. [New NCERT Class 11th Page No. 213] 181. (4) NaCl is absent in sebum. [New NCERT Class 11th Page No. 213] 182. (4) Glomerular filtration pressure in a normal healthy man is around 125 mm Hg/min. [New NCERT Class 11th Page No. 209] 183. (1) Glucose, amino acid, Na⁺ etc in the filtrate are reabsorbed by active transport. [New NCERT Class 11th Page No. 206] 184. (3) JGA is a special sensitive region formed by cellular modifications in the distal convoluted tubule and the afferent arteriole at the location of their contact. [Old NCERT Class 11th Page No. 294] 185. (1) A hairpin-shaped Henle's loop has a descending limb, followed by an ascending limb. [Old NCERT Class 11th Page No. 293] 186. (1) Dialysing fluid used in hemodialysis has a similar composition to plasma but without nitrogenous wastes. It contains electrolytes and heparin to prevent blood clotting during the dialysis process, which facilitates the removal of waste products from the blood. [Old NCERT Class 11th Page No. 294]
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187. (2)

Ammonia is the most toxic form of nitrogenous waste and requires large amount of water for its elimination, whereas uric acid, being the least toxic, can be removed with a minimum loss of water.

[New NCERT Class 11th Page No. 205]

188. (2)



[New NCERT Class 11th Page No. 207]

189. (1)

Erythropoietin hormone is released by Juxtaglomerular apparatus.

[New NCERT Class 11th Page No. 202]

190. (3)

Aldosterone hormone leads to reabsorption of Na^+ and water from the distal part of renal tubule.

[New NCERT Class 11th Page No. 212]

191. (2)

- ADH facilitates water reabsorption from latter parts of the tubule, thereby preventing diuresis.
- During summer when body loses lot of water by evaporation, the release of ADH is increased.

[New NCERT Class 11th Page No. 212]

192. (1)

Hydrogen ion are actively secreted into the filtrate.

[New NCERT Class 11th Page No. 213]

193. (3)

A healthy human being excrete 25-30gm of urea per day.

[New NCERT Class 11th Page No. 213]

194. (1)

Oxalate salt is involved in the formation of renal calculi.

[New NCERT Class 11th Page No. 214]

195. (4)

Ultrafiltration	Malpighian corpuscle
Concentration of urine	Henle's loop
Transport of urine	Ureter
Storage of urine	Urinary bladder

[New NCERT Class 11th Page No. 214]

196. (3)

Glomerulonephritis is the inflammation of the glomeruli.

[New NCERT Class 11th Page No. 214]

197. (4)

On an average, 1100 – 1200 ml of blood is filtered by the kidneys per minute which constitute roughly $\frac{1}{5}$ th of the blood pumped out by each ventricle of the heart in a minute.

[New NCERT Class 11th Page No. 208]

198. (3)

Vasa recta is absent or highly reduced in cortical nephrons.

[New NCERT Class 11th Page No. 208]

199. (3)

Urethral sphincter cause the release of urine.

[New NCERT Class 11th Page No. 213]

200. (1)

Urine is stored in the urinary bladder until a voluntary signal from the CNS.

[New NCERT Class 11th Page No. 215]

